ture, data centres, compute capacity and clean power. We also need to invest across the value chain in other frontier technologies such as nuclear fission and fusion. In parallel, as studies highlight, India needs to raise annual incremental climate and energy transition investment by roughly 2 per cent of GDP, or about $80-85 billion (Rs 7.2-7.7 lakh crore) in 2025 dollar terms.

India's financial system will have to be fundamentally redesigned to massively increase corporate capital expenditures. Much of India's financial architecture is geared toward short- to medium-duration lending, asset-backed finance and risk minimisation. That structure supported earlier phases of development focused on macroeconomic stability and basic infrastructure. We now need to pivot to a financial system that can finance high levels of technology risk, long gestation periods and irreversible capital commitments. Simply put, our investment managers have to be skilled enough to lose billions of dollars on individual technology bets but still make excellent returns through good portfolio construction.

Global capital, while important, cannot be the primary driver. Today, global investors are deploying capital in advanced economies where returns are attractive and risks are familiar. The surge in AI infrastructure investment in the US illustrates this clearly. Global capital can earn high returns in stable jurisdictions with deep markets and predictable regulation. We need to understand these realities.

To compete, we must deepen our pools of long-term domestic capital. This capital is more stable during periods of global volatility and is naturally aligned with national development priorities. Household financial savings are rising while pension and insurance assets are also expanding quickly. Our capital markets have matured significantly. Yet a large share of this capital remains locked into short-duration or low-risk instruments. We must modify alternative investment fund (AIF) structures to enable long-term investment into frontier technologies and sustainable infrastructure.

TAKEAWAYS

- **To compete globally, India must massively fund frontier tech. Also gumption to back intentional technological risks**
- **US commits $500 bn yearly to AI, China invests $100 bn; India's AI Mission: $1 bn over 5 years ($200 mn annually)**
- **Capital markets are now mature, must enable long-term investment in frontier tech, sustainable infrastructure**

COUNTRIES THAT **FAIL TO FINANCE THEIR OWN FRONTIER TECHNOLOGIES** WILL BECOME PRICE TAKERS IN SYSTEMS DESIGNED ELSEWHERE. **THIS CAN'T BE INDIA'S FUTURE**

The AI build-out illustrates the stakes with particular clarity. India's success with digital public infrastructure such as Aadhaar, UPI and the Account Aggregator framework shows that population-scale technology systems must be embedded within sovereign governance structures. AI represents the next layer of the digital economy, shaping education, healthcare, finance, logistics and public administration. If the intelligence layer is owned, trained and governed elsewhere, oversight weakens, and accountability erodes.

Our current investments in AI fall well short of what is required for global leadership. The government's IndiaAI Mission has been approved at about a billion dollars over five years; we will probably deploy just $200 million (Rs 1,803 crore) per year. India's largest technology firms and infrastructure players are investing more. TCS, alongside PE firm TPG (Texas Pacific Group), is building multi-billion-dollar AI data centre capacity. Microsoft, Google and Amazon have each announced long-term investments running into tens of billions of dollars to expand cloud and AI infrastructure in India. Reliance has unveiled an ambitious plan to build gigawatt scale AI compute capacity, while Adani is investing heavily in hyperscale data centres. Collectively, experts estimate that this adds up to about $10 billion (Rs 90,000 crore) annually.

This is how techno-colonialism may manifest itself in the 21st century. Most countries will not be able to invest sufficiently in building up their digital infrastructure. Global powers will foster dependence on external compute infrastructure, proprietary models, closed standard and capital structures beyond domestic regulatory reach. Countries that fail to finance their own frontier technologies will become price takers in systems designed elsewhere. This cannot be India's future.

A truly Samriddh Bharat will be achieved when we combine Viksit and Harit Bharat. But reaching this green frontier will require India to redesign its financial system to take risks intentionally, share that risk between public and private balance sheets, and mobilise large pools of long-term domestic capital. Our ambition will be fulfilled only when we make these structural shifts to our financial system. ■

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